Search System Redesign

The current search system retrieves products using keyword matching on product name and attributes, then ranks by text relevance, popularity, and business rules. After analyzing 255K query sessions and running a systematic modeling effort in Part 2, the central finding is that the performance bottleneck is retrieval, not ranking. 68.7% of sessions produce zero clicks. Among sessions that do produce clicks, the current system already ranks well. The click model could not beat the existing ordering despite testing three model architectures, index debiasing method, and many feature combinations. Hence, the opportunity is in getting relevant products into the result set in the first place. Before I delve into the suggested redesign, I want to highlight key findings that drove my decision making.

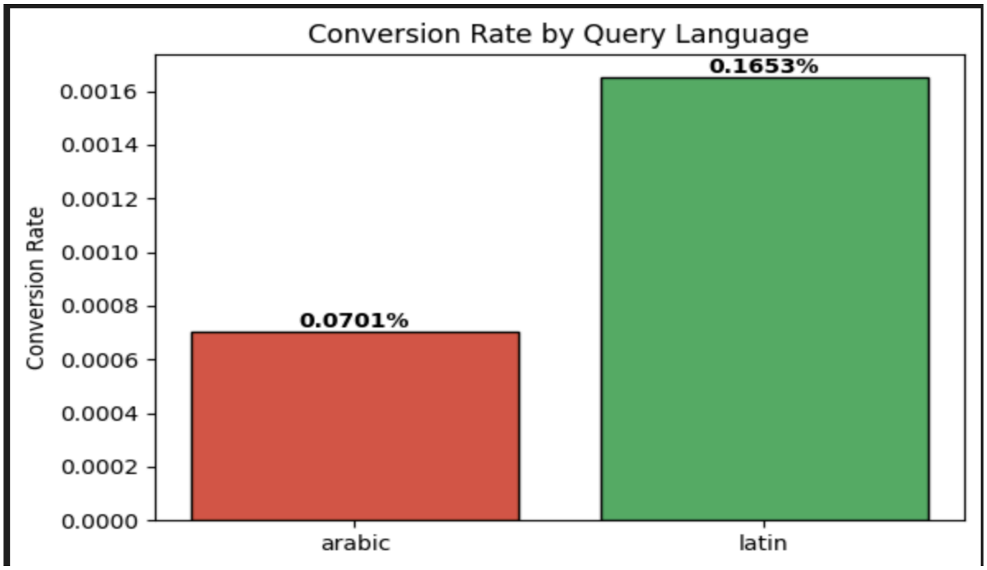
Three distinct failure patterns

These patterns are important to flag because they help define the problem space and consequently the possible solution set.

The three distinct failure patterns drive the zero-click rate (each pointing to a different retrieval gap):

Pattern 1: Arabic brand transliterations. 40% of queries are in Arabic while 99% of the catalog is in English. High-intent navigational queries like غوتشي (Gucci, 69% failure) and شانيل (Chanel) fail because keyword matching cannot bridge Arabic script to Latin catalog entries.

These users know exactly what they want but the system can't understand the query. Arabic and Latin queries show comparable CTR (~2%), but Arabic converts at half the rate (1% vs 2%), suggesting that when Arabic retrieval does work, it surfaces less relevant results.



Pattern 2: Broad category queries. Exploratory queries like "Evening gowns" (99.8% failure), "Perfume" (91%), and "White shirt" (91%) fail across both languages. The system retrieves products but nothing compelling enough to click.

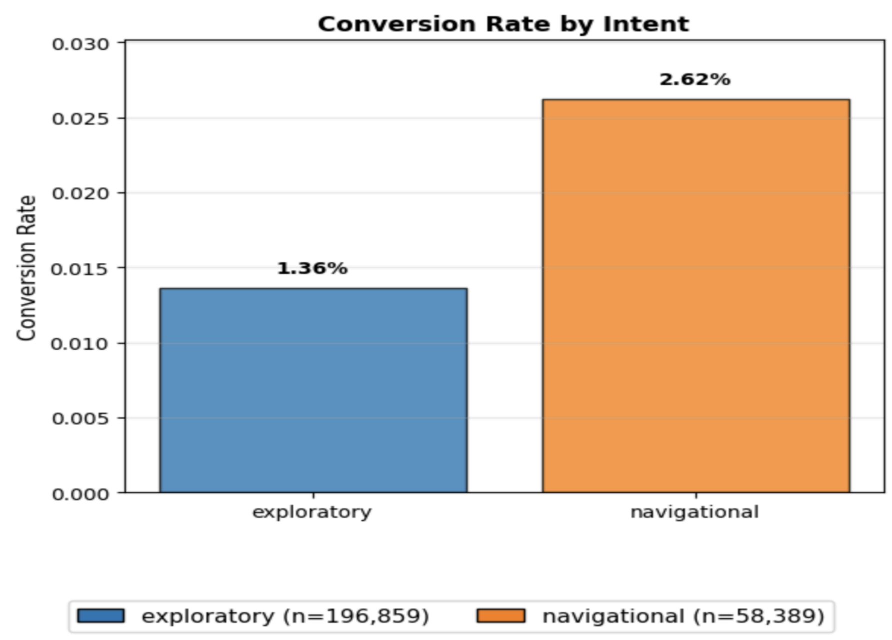
Pattern 3: Specific long-tail queries. Near-exact product descriptions like "Dolce & Gabbana D&G Slides in Rubber" (87% failure) and "Charlotte Tilbury Pillow Talk" (100% failure) break keyword matching because it is possible that multi-token queries fragment and dilute precision. The more specific the user is, the worse the system performs although this is the opposite of what should happen.

An interesting realization here is that the use of crosslingual (arabic/english) sentence embeddings can address these 3 patterns. We'll delve more into that later in this document.

Other key findings:

~5% of impressions are tied to products with missing metadata (no designer, no class), contributing zero conversions. These catalog ghosts consume result slots without providing value.

Navigational queries (brand-specific) convert at 2x the rate of exploratory queries when they succeed, building the case that intent type should drive retrieval strategy: precision for navigational, breadth for exploratory.



These findings inform the following redesign, ordered from quick operational fixes to architectural changes.

Optimization Sequence

Catalog Cleanup: In the current dataset ~5% of impressions tied are to products with missing metadata (no designer, no class, zero conversions). Ensure catalog ingestion validates completeness before products enter the search index.

Arabic brand transliteration mapping: Arabic and Latin queries show comparable CTR (~2%), but Latin converts at 2.4x Arabic (0.16% vs 0.07%).

A key driver is transliteration mismatch: غوتشي (Gucci), شانيل (Chanel), كوتش (Coach) fail to match their English catalog entries. A phonetic lookup table mapping the top 50–100 luxury brand names to their common Arabic transliterations resolves the majority of these failures.

Here’s a sample for demonstration:

Brand (Latin)	Arabic Transliteration
Gucci	غوتشي

Chanel	شانيل
Coach	كوتش
Louis Vuitton	لويس فويتون
Dior	ديور
Prada	برادا
Hermès	هيرميس
Dolce & Gabbana	دولتشي آند غابانا
Balenciaga	بالنسياغا
Versace	فيرساتشي

Key-matching can check if the arabic keyword matches any of the transliterations. Maintaining this list, even if only for the top selling brands, can provide a quick win for a gap the current system cannot handle. You can make this solution more robust by adding fuzzy matching (e.g. levenshtein distance) such that if an arabic keyword for Brand Gucci like كوتشي still gets mapped to غوتشي.

Intent router: A binary classifier distinguishing navigational (brand-seeking) from exploratory (category-browsing) queries. As we've seen earlier, navigational queries (brand-specific) lead to higher conversion. This means we should utilize a retrieval pipeline that emphasizes precision over recall. Generic exploratory queries have lower conversions, meaning there is an opportunity here to utilize a retrieval pipeline that prioritizes recall/coverage.

As for implementation, we can start with a simple substring matching approach against designer names we already know. The more sophisticated approach would be to build a binary classifier that tags the query accordingly.

This should be early in the pipeline because it determines which downstream components activate (making the overall system more efficient).

Query-to-category classifier:
A small classifier that takes the search query as input and predicts the relevant product class (Dresses, Bags, Shoes, etc.). This constrains the retrieval space before search executes, so instead of matching "evening gowns" against the entire catalog, retrieval is scoped to the Dresses class. This reduces false positives (irrelevant products consuming result slots) and false negatives (relevant products buried by broad matching). It also enables more efficient catalog indexing by product class.

Cross-lingual semantic embeddings:
Replace pure keyword matching with multilingual embedding-based retrieval (e.g., multilingual sentence transformers). Queries and product metadata (name + designer + class) are embedded into a shared vector space where semantic similarity and cross-lingual matching work. Arabic queries match English product names without transliteration. This is the highest-impact architectural change but requires embedding infrastructure (model serving, vector index, periodic re-indexing).

We will connect the dots between these optimizations in the following section.

Suggested implementation

The proposed system routes every query through three stages:

First, language detection and intent classification determine whether the query is navigational or exploratory. Navigational queries pass through a transliteration normalization step (mapping Arabic brand names to their English equivalents), then retrieve from a brand-filtered index using cross-lingual embeddings for precision. Exploratory queries pass through a query-to-category classifier that constrains the search space to the relevant product class, then retrieve using hybrid search (BM25 keyword matching combined with semantic embeddings) for broader recall.

Both paths converge at a click model ranker that re-orders the retrieved candidates before serving final results.

```
Query → Language Detection → Intent Router
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                    ┌─────────────┴─────────────┐
                    │                           │
               Navigational                Exploratory
                    │                           │
          Transliteration              Query-to-Category
             Normalization                 Classifier
                    │                           │
```

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     Brand-filtered              Category-scoped
   Embedding Retrieval         Hybrid Retrieval
            │                  (BM25 + Embeddings)
            │                          │
            └─────────────┬────────────┘
                          │
                  Click Model Ranker
                          │
                    Final Results
```

How components of the suggested redesign address initial findings:

Finding	Component that addresses it
68.7% zero-click sessions	Hybrid retrieval + embeddings (retrieval, not ranking)
Arabic 2.4x lower conversion	Transliteration mapping + cross-lingual embeddings
"Evening gowns" 99.8% failure rate	Semantic retrieval for exploratory queries
Navigational 2x conversion but lower success	Intent router + brand-filtered precision retrieval
5% catalog ghosts	Catalog cleanup at ingestion